

Purushotham Vadde

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 9+ years building distributed systems, high-availability microservices, and cloud-native data pipelines. At Adobe, leading AI-assisted development of the Adobe Journey Optimizer (AJO) B2B platform using Claude Code. Delivered systems processing 10TB+ monthly, serving 10,000+ daily users, and cutting deployment cycles from 4 hours to 20 minutes across Java, Spring Boot, Spark, Databricks, AWS, and Azure.

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Scala, Python, FastAPI, RESTful APIs

System Design: Distributed Systems, Microservices, High Availability, Event-Driven Architecture

Data Engineering: Spark, Databricks, Kafka, Airflow, Snowflake, Delta Lake, ETL/ELT

Cloud & Databases: AWS, Azure, MySQL, MongoDB, Redis, CosmosDB

DevOps & CI/CD: Docker, Kubernetes, GitHub Actions, Argo CD, Liquibase

Monitoring & Observability: Grafana, Prometheus, Splunk

AI & ML: Claude Code, LLMs, RAG, Prompt Engineering, NLP

WORK EXPERIENCE

Senior Software Engineer

Jan 2026 – Present

Adobe, Inc. | Denver, CO

- **AI-Assisted Development:** Designed and developed AJO B2B Edition backend using Claude Code for scaffolding, test generation, and pattern enforcement; cut boilerplate time by ~40% and delivered 5 production services in Q1.
- **Metadata Service:** Built core metadata layer (Spring Boot, MySQL, Netflix Hollow) for AJO B2B journey orchestration; Hollow producer publishes per-tenant delta snapshots to S3, consumed into an in-memory state engine for sub-millisecond reads.
- **Kafka Consumer Service:** Built a fault-tolerant Kafka consumer (Spring Kafka) with idempotent handlers, retry policies, and DLQ routing; processes 500K+ events/day at 99.9% ingestion reliability with zero message loss.
- **Data Ingestion Pipeline:** Engineered incremental ETL (Spark, Databricks) from Snowflake into AEP Delta Lakehouse with schema evolution and automated quality validation; reduced data freshness latency from 24h to under 2h.
- **Schema Management:** Implemented Liquibase DDL versioning across 10+ microservices with automated rollback strategies, achieving zero-downtime releases.

Software Engineer

Sep 2021 – Dec 2025

Adobe, Inc. | Denver, CO

- **AEP Analytics Dashboard:** Built accounts and persons analytics dashboard on AEP; ingested MySQL data, authored Spark/Python jobs to compute aggregated metrics written to Azure Synapse Analytics, and

automated the pipeline with Airflow; cut report refresh latency to under 30 minutes.

- **Batch ETL Pipelines:** Engineered Spark/Databricks jobs processing 10TB+ monthly from S3 to Azure Synapse via Airflow with partition pruning and broadcast joins; maintained 99.5% data accuracy.
- **Dynamic Chat Platform:** Built a real-time WebSocket chat platform with Redis pub/sub for near-zero-latency messaging, Java WebPush for agent lifecycle notifications, and Spring Boot microservices; served 10,000+ daily users at 99.9% uptime and <200ms response time.
- **CI/CD & Observability:** Built CI/CD pipelines (Ethos Flex, Moonbeam, Argo CD) cutting deploy time from 4h to 20 min; implemented distributed tracing and alerting with Grafana, Prometheus, and Splunk across 50+ microservices.

Software Engineer

May 2015 – Jul 2019

Cognizant Technology Solutions | India

- **REST API Backend:** Built a Python REST API for a Fortune 500 insurance claims platform with RBAC and PII compliance; supported 200+ concurrent users.
- **PySpark ETL Pipelines:** Processed 5TB+ monthly from MySQL, MongoDB, S3, and SFTP on AWS EMR; achieved 2.5x performance gain via adaptive partitioning and Spark SQL optimization at 99.7% success rate.
- **Redshift Data Warehouse:** Designed star schema and medallion architecture for 4x query improvement; automated S3-to-Redshift loading via Lambda with Parquet compression, cutting storage costs by 35%.

EDUCATION

Doctoral Certificate in Business Administration

Jul 2024 – Aug 2025

Westcliff University

Master of Science in Data Science & Analytics

Aug 2019 – May 2021

University of Oklahoma